EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. 6 married couples are standing in a room. If 4 people are chosen at random, then the chance that exactly one married couple is among the 4 is-

(A) $\frac{16}{33}$ (B) $\frac{8}{33}$ (C) $\frac{17}{33}$ (D) $\frac{24}{33}$

MPB164

2. The probability that a positive two digit number selected at random has its tens digit at least three more than its unit digit is -

(A) 14/45 (B) 7/45 (C) 36/45 (D) 1/6

MPB165

3. A 5 digit number is formed by using the digits 0,1,2,3,4 & 5 without repetition. The probability that the number is divisible by 6 is :

(A) 8% (B) 17% (C) 18% (D) 36%

MPB166

4. Mr. A forgot to write down a very important phone number. All he remembers is that it started with 713 and that the next set of 4 digit involved are 1, 7 and 9 with one of these numbers appearing twice. He guesses a phone number and dials randomly. The odds in favour of dialing the correct telephone number, is -

(A) 1 : 35 (B) 1 : 71 (C) 1 : 23 (D) 1 : 36

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MPB167

5. There are 4 defective items in a lot consisting of 10 items. From this lot we select 5 items at random. The probability that there will be 2 defective items among them is -

(A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{5}{21}$ (D) $\frac{10}{21}$

MPB168

6. If two of the 64 squares are chosen at random on a chess board, the probability that they have a side in common is -

(A) 1/9 (B) 1/18 (C) 2/7 (D) none

MPB169

7. Two red counters, three green counters and 4 blue counters are placed in a row in random order. The probability that no two blue counters are adjacent is -

(A) $\frac{7}{99}$ (B) $\frac{7}{198}$ (C) $\frac{5}{42}$ (D) none

MPB170

8. South African cricket captain lost the toss of a coin 13 times out of 14. The chance of this happening was

(A) $\frac{7}{2^{13}}$ (B) $\frac{1}{2^{13}}$ (C) $\frac{13}{2^{14}}$ (D) $\frac{13}{2^{13}}$

MPB171

For More Join: @IITJEE_Advanced

9. There are ten prizes, five A's, three B's and two C's, placed in identical sealed envelopes for the top ten contestants in a mathematics contest. The prizes are awarded by allowing winners to select an envelope at random from those remaining. When the 8th contestant goes to select the prize, the probability that the remaining three prizes are one A, one B and one C, is

(A) 1/4 (B) 1/3 (C) 1/12 (D) 1/10

MPB172

10. A coin is tossed and a die is thrown, then the probability that the outcome will be a head or a number greater than 4.

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

MPB173

11. A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant chosen has the value non negative is :

(A) 3/16 (B) 6/16 (C) 10/16 (D) 13/16

MPB174

12. A card is drawn & replaced in an ordinary pack of 52 playing cards. Minimum number of times must a card be drawn so that there is atleast an even chance of drawing a heart, is

(A) 2 (B) 3 (C) 4 (D) more than four

MPB175

13. A license plate is 3 capital letters (of English alphabets) followed by 3 digits. If all possible license plates are equally likely, the probability that a plate has either a letter palindrome or a digit palindrome (or both), is

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(A) $\frac{7}{52}$ (B) $\frac{9}{65}$ (C) $\frac{8}{65}$ (D) none

MPB176

14. Whenever horses a, b, c race together, their respective probabilities of winning the race are 0.3, 0.5 and 0.2 respectively. If they race three times the probability that "the same horse wins all the three races" and the probability that a, b, c each wins one race, are respectively (Assume no dead heat)

(A) $\frac{8}{50}$; $\frac{9}{50}$ (B) $\frac{16}{100}$, $\frac{3}{100}$ (C) $\frac{12}{50}$; $\frac{15}{50}$ (D) $\frac{10}{50}$; $\frac{8}{50}$

MPB177

15. Two cubes have their faces painted either red or blue. The first cube has five red faces and one blue face. When the two cubes are rolled simultaneously, the probability that the two top faces show the same colour is 1/2. Number of red faces on the second cube, is

(A) 1 (B) 2 (C) 3 (D) 4

MPB178

16. A committee of three persons is to be randomly selected from a group of three men and two women and the chair person will be randomly selected from the committee. The probability that the committee will have exactly two women and one man, and that the chair person will be a woman, is/are

(A) 1/5 (B) 8/15 (C) 2/3 (D) 3/10

MPB179

For More Join: @IITJEE_Advanced

17. A bag contains 3 R & 3 G balls and a person draws out 3 at random. He then drops 3 blue balls into the bag & again draws out 3 at random. The chance that the 3 later balls being all of different colours is

(A) 15% (B) 20% (C) 27% (D) 40%

MPB180

18. An urn contains 10 balls coloured either black or red. When selecting two balls from the urn at random, the probability that a ball of each colour is selected is $8/15$. Assuming that the urn contains more black balls than red balls, the probability that at least one black ball is selected, when selecting two balls, is

(A) $\frac{18}{45}$ (B) $\frac{30}{45}$ (C) $\frac{39}{45}$ (D) $\frac{41}{45}$

MPB181

19. The number 'a' is randomly selected from the set {0, 1, 2, 3, ... 98, 99}. The number 'b' is selected from the same set. Probability that the number $3^a + 7^b$ has a digit equal to 8 at the units place, is

(A) $\frac{1}{16}$ (B) $\frac{2}{16}$ (C) $\frac{4}{16}$ (D) $\frac{3}{16}$

MPB182

20. If a, b and c are three numbers (not necessarily different) chosen randomly and with replacement from the set {1, 2, 3, 4, 5}, the probability that $(ab + c)$ is even, is

(A) $\frac{35}{125}$ (B) $\frac{59}{125}$ (C) $\frac{64}{125}$ (D) $\frac{75}{125}$

MPB183

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21. An ant is situated at the vertex A of the triangle ABC. Every movement of the ant consists of moving to one of other two adjacent vertices from the vertex where it is situated. The probability of going to any of the other two adjacent vertices of the triangle is equal. The probability that at the end of the fourth movement the ant will be back to the vertex A, is :

(A) 4/16 (B) 6/16 (C) 7/16 (D) 8/16

MPB184

22. Mr. A and Mr. B each have a bag that contains one ball of each of the colours blue, green, orange, red and violet. 'A' randomly selects one ball from his bag and puts it into B's bag. 'B' then randomly selects one ball from his bag and puts it into A's bag. The probability that after this process the contents of the two bags are the same, is

(A) 1/6 (B) 1/5 (C) 1/3 (D) 1/2

MPB185

23. A box contains 10 tickets numbered from 1 to 10. Two tickets are drawn one by one without replacement. The probability that the "absolute value of difference between the first drawn ticket number and the second is not less than 4" is

(A) $\frac{7}{30}$ (B) $\frac{7}{15}$ (C) $\frac{11}{30}$ (D) $\frac{10}{30}$

MPB186

24. The probability that a radar will detect an object in one cycle is p. The probability that the object will be detected in n cycles is :

(A) $1 - p^n$ (B) $1 - (1 - p)^n$ (C) p^n (D) $p(1 - p)^{n-1}$

MPB187

For More Join: @IITJEE_Advanced

25. Three numbers are chosen at random without replacement from {1, 2, 3,....., 10}. The probability that the minimum of the chosen numbers is 3 or their maximum is 7 is

(A) 1/2 (B) 1/3 (C) 1/4 (D) 11/40

MPB188

26. In a maths paper there are 3 sections A, B & C. Section A is compulsory. Out of sections B & C a student has to attempt any one. Passing in the paper means passing in A & passing in B or C. The probability of the student passing in A, B & C are p, q & $1/2$ respectively student select randmoly one of the option out of B & C. If the probability that the student is successful is $1/2$ then possible choosen can be :

(A) $p = q = 1$ (B) $p = q = ½$ (C) $p = 1, q = 0$ (D) $p = 1, q = ½$

MPB189

27. If C and D are two events such that $C \subset D$ and $P(D) \neq 0$, then the correct statement among the following is :-

(A) $P(C|D) < P(C)$ (B) $P(C|D) = \frac{P(D)}{P(C)}$ (C) $P(C|D) = P(C)$ (D) $P(C|D) \geq P(C)$

MPB190

28. Consider 5 independent Bernoulli's trials each with probability of success p. If the probability of at least one failure is greater than or equal to $\frac{31}{32}$, then p lies in the interval :-

(A) $\left[0, \frac{1}{2}\right]$ (B) $\left(\frac{11}{12}, 1\right]$ (C) $\left(\frac{1}{2}, \frac{3}{4}\right]$ (D) $\left(\frac{3}{4}, \frac{11}{12}\right]$

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MPB191

29. Let A, B, C be pairwise independent events with $P(C) > 0$ and $P(A \cap B \cap C) = 0$. Then $P(A^c \cap B^c | C)$ is equal to :

(A) $P(A^c) - P(B)$ (B) $P(A) - P(B^c)$ (C) $P(A^c) + P(B^c)$ (D) $P(A^c) - P(B^c)$

MPB192

30. Three numbers are chosen at random without replacement from {1, 2, 3, ..., 8}. The probability that their minimum is 3, given that their maximum is 6, is :

(A) $\frac{2}{5}$ (B) $\frac{3}{8}$ (C) $\frac{1}{5}$ (D) $\frac{1}{4}$

MPB193

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. A boy has a collection of blue and green marbles. The number of blue marbles belong to the sets {2, 3, 4,..... 13}. If two marbles are chosen simultaneously and at random from his collection, then the probability that they have different colour is 1/2. Possible number of blue marbles is :

(A) 2 (B) 3 (C) 6 (D) 10

MPB194

2. If A & B are two events such that $P(B) \neq 1$, B^C denotes the event complementary to B, then

(A) $P\left(A/B^C\right) = \dfrac{P(A) - P(A \cap B)}{1 - P(B)}$

(B) $P(A \cap B) \geq P(A) + P(B) - 1$

(C) $P(A) > < P\left(A/B\right)$ according as $P\left(A/B^C\right) > < P(A)$

(D) $P\left(A/B^C\right) + P\left(A^C/B^C\right) = 1$

MPB195

3. For $P(A) = \dfrac{3}{8}$; $P(B) = \dfrac{1}{2}$; $P(A \cup B) = \dfrac{5}{8}$ which of the following do/does hold good?

(A) $P\left(A^c/B\right) = 2P\left(A/B^c\right)$ (B) $P(B) = P\left(A/B\right)$

(C) $15P\left(A^c/B^c\right) = 8P\left(B/A^c\right)$ (D) $P\left(A/B^c\right) = P(A \cap B)$

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MPB196

4. If E_1 and E_2 are two events such that $P(E_1) = 1/4$, $P(E_2/E_1) = 1/2$ and $P(E_1/E_2) = 1/4$

(A) then E_1 and E_2 are independent

(B) E_1 and E_2 are exhaustive

(C) E_2 is twice as likely to occur as E_1

(D) Probabilities of the events $E_1 \cap E_2$, E_1 and E_2 are in G.P.

MPB197

5. Two events A and B are such that the probability that at least one of them occurs is 5/6 and both of them occurring simultaneously is 1/3. If the probability of not occurrence of B is 1/2 then

(A) A and B are equally likely (B) A and B are independent

(C) $P(A/B) = 2/3$ (D) $3\,P(A) = 4\,P(B)$

MPB198

6. The probabilities of events, $A \cap B$, A, B & $A \cup B$ are respectively in A.P. with probability of second term equal to the common difference. Therefore the events A and B are

(A) mutually exclusive

(B) independent

(C) such that one of them must occur

(D) such that one is twice as likely as the other

MPB199

For More Join: @IITJEE_Advanced

7. A box contains 11 tickets numbered from 1 to 11. Six tickets are drawn simultaneously at random. Let E_1 denotes the event that the sum of the numbers on the tickets drawn is even and E_2 denotes the event that the sum of the numbers on the tickets drawn is odd. Which of the following hold good?

(A) E_1 and E_2 are equally likely

(B) E_1 and E_2 are exhaustive

(C) $P(E_2) > P(E_1)$

(D) $P(E_1/E_2) = P(E_2/E_1)$

MPB200

8. If $\bar{E}$ & $\bar{F}$ are the complementary events of events E & F respectively & if $0 < P(F) < 1$, then :

(A) $P(E|F) + P(\bar{E}|F) = 1$

(B) $P(E|F) + P(E|\bar{F}) = 1$

(C) $P(\bar{E}|F) + P(E|\bar{F}) = 1$

(D) $P(E|\bar{F}) + P(\bar{E}|\bar{F}) = 1$

MPB201

9. Probability of $\boldsymbol{n}$ heads in $\boldsymbol{2n}$ tosses of a fair coin can be given by

(A) $\prod_{r=1}^{n}\left(\frac{2r-1}{2r}\right)$

(B) $\prod_{r=1}^{n}\left(\frac{n+r}{2r}\right)$

(C) $\sum_{r=0}^{n}\left(\frac{{}^nC_r}{2^n}\right)^2$

(D) $\frac{\sum_{r=0}^{n}\left({}^nC_r\right)^2}{\left(\sum_{r=0}^{n}{}^nC_r\right)^2}$

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MPB202

10. Which of the following statements is/are True?

(A) A fair coin is tossed n times where n is a positive integer. The probability that n^{th} toss results in head is $1/2$.

(B) The conditional probability that the nth toss results in head given that first $(n-1)$ tosses results in head is $1/2^n$.

(C) Let E and F be the events such that F is neither impossible nor sure. If $P(E/F) > P(E)$ then $P(E/F^c) > P(E)$.

(D) If A, B and C are independent then the events $(A \cup B)$ and C are independent.

MPB203

11. Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If $P(T)$ denotes the probability of occurrence of the event T, then -

(A) $P(E)=\frac{4}{5}, P(F)=\frac{3}{5}$

(B) $P(E)=\frac{1}{5}, P(F)=\frac{2}{5}$

(C) $P(E)=\frac{2}{5}, P(F)=\frac{1}{5}$

(D) $P(E)=\frac{3}{5}, P(F)=\frac{4}{5}$

MPB204

For More Join: @IITJEE_Advanced

12. A ship is fitted with three engines E_1, E_2 and E_3. The engines function independently of each other with respective probabilities $\frac{1}{2}, \frac{1}{4}$ and $\frac{1}{4}$. For the ship to be operational at least two of its engines must function. Let X denote the event that the ship is operational and X_1, X_2, X_3 denotes respectively the events that the engines E_1, E_2 and E_3 are functioning. Which of the following is (are) true?

(A) $P\left[X_1^c \mid X\right]=\frac{3}{16}$

(B) P[Exactly two engines of ship are functioning $\mid X]=\frac{7}{8}$

(C) $P[X \mid X_2]=\frac{5}{16}$

(D) $P[X \mid X_1]=\frac{7}{16}$

MPB205

13. Let X and Y be two events such that $P(X \mid Y)=\frac{1}{2}, P(Y \mid X)=\frac{1}{3}$ and $P(X \cap Y)=\frac{1}{6}$. Which of the following is(are) correct ?

(A) $P(X \cup Y)=\frac{2}{3}$ (B) X and Y are independent

(C) X and Y are not independent (D) $P\left(X^C \cap Y\right)=\frac{1}{3}$

Rankers Academy JEE

MPB206

14. A lot contains 50 defective & 50 non defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as :

A = {the first bulb is defective}; B = {the second bulb is non defective}

C = {the two bulbs are both defective or both non defective}. Then which of the following is/are correct.

(A) A, B, C are pair wise independent (B) A, B, C are independent

(C) A, B, C are not pair wise independent (D) A,B,C are not independent

MPB207

15. A bomber wants to destroy a bridge. Two bombs are sufficient to destroy it. If four bombs are dropped, if the chance of a bomb hitting the target is 0.4. Then the probability that it is destroyed, is less than

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$

MPB208

16. Before a race the chance of three runners, A, B, C were estimated to be proportional to 5, 3, 2, but during the race A meets with an accident which reduces his chance to 1/3. Now $P(B)$ and $P(C)$ denotes the respective probabilities for winning B & C, then

(A) $P(B)=\frac{2}{5}$ (B) $P(C)=\frac{4}{15}$ (C) $P(B)=\frac{1}{4}$ (D) $P(C)=\frac{1}{6}$

MPB209

For More Join: @IITJEE_Advanced

17. A box contains 11 tickets numbered from 1 to 11. Two tickets are drawn simultaneously at random. Let E_1 denotes the event that the sum of the numbers on the tickets drawn is even and E_2 denotes the event that the sum of the numbers on the tickets drawn is odd
Which of the following hold good ?
(A) $P(E_1/E_2) = P(E_2/E_1)$ (B) E_1 and E_2 are exhaustive
(C) $P(E_2) > P(E_1)$ (D) E_1 and E_2 are equally likely

MPB210

18. In an experimental performance of a single throw of a pair of unbiased normal dice, three events E_1, E_2 & E_3 are defined as follows :
E_1 : getting a prime numbered face on each dice
E_2 : getting the same number on each dice
E_3 : getting a sum total of dots on two dice equal to 8. Then :
(A) the events E_1, E_2 & E_3 are mutually exclusive
(B) the events E_1, E_2 & E_3 are not pairwise mutually exclusive
(C) the events E_1, E_2 are independent
(D) $P(E_3 \mid E_1) = 2/9$.

MPB211

19. The probability that a bulb produced by a factory will fuse after an year of use is 0.1. Then the probability that out of 4 such bulbs

(A) None of then bulb will fuse after an year of use is $\frac{9^4}{10^4}$

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(B) More then three bulbs will fuse after an year of use is $\frac{1}{10^4}$

(C) Not more then three bulbs will fuse after an year of use is $\frac{9999}{10000}$

(D) All the bulbs will fuse after an year of use is $\frac{1}{10^4}$

MPB212

20. An unbiased coin is tossed n times. Let X denote the number of times head occurs. If $P(X = 4)$, $P(X = 5)$ and $P(X = 6)$ are in AP, then the value of n can be
(A) 7 (B) 10 (C) 12 (D) 14

MPB213

21. Ram post a letter to Ghanshyam. It is known that one letter out of 10 letters do not reach its destination. If it is certain that Ghanshyam will reply if he receives the letter. If A denotes the event that Ghanshyam receives the letter and B that Ram gets a reply, then -

(A) $P(B) = \frac{81}{100}$ (B) $P(A \cap B) = \frac{81}{100}$

(C) $P\left(\frac{A}{\overline{B}}\right) = \frac{9}{19}$ (D) $P(A \cup B) = \frac{9}{10}$

MPB214

For More Join: @IITJEE_Advanced

22. Before a race the chance of three runners A, B & C were estimated to be proportional to 5, 3 & 2 respectively but during the race A meets with an accident which reduces his chance to 1/3. If the respective chances of B and C are $P(B)$ and $P(C)$ then -

(A) $P(B) = \frac{2}{5}$ (B) $P(C) = \frac{4}{15}$

(C) $P(C) = \frac{2}{5}$ (D) $P(B) = \frac{4}{15}$

MPB215

23. Two real numbers, x & y are selected at random. Given that $0 \le x \le 1;\ 0 \le y \le 1$. Let A be the event that $y^2 \le x$; B be the event that $x^2 \le y$, then-

(A) $P(A \cap B) = \frac{1}{3}$

(B) A & B are exhaustive events

(C) A & B are mutually exclusive

(D) A & B are independent events.

MPB216

24. Three dice are rolled simultaneously, then the chance that exactly two of the dice show the same number is more than -

(A) $\frac{5}{18}$ (B) $\frac{5}{9}$ (C) $\frac{5}{13}$ (D) $\frac{1}{3}$

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MPB217

25. Suppose that M is a natural number with the property that if x is chosen randomly from the set $\{1, 2, \ldots\ldots, 1000\}$ the probability that x is divisor of M is $\frac{1}{2}$. If $M \le 1000$ then maximum possible value of M is divisible by -

(A) 2 (B) 61 (C) 3 (D) 5

MPB218

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 26 to 27

A JEE aspirant estimates that she will be successful with an 80 percent chance if she studies 10 hours per day, with a 60 percent chance if she studies 7 hours per day and with a 40 percent chance if she studies 4 hours per day. She further believes that she will study 10 hours, 7 hours and 4 hours per day with probabilities 0.1, 0.2 and 0.7, respectively

26. The chance she will be successful, is

(A) 0.28 (B) 0.38 (C) 0.48 (D) 0.58

MPB219

27. Given that she is successful, the chance she studied for 4 hours, is

(A) $\frac{6}{12}$ (B) $\frac{7}{12}$ (C) $\frac{8}{12}$ (D) $\frac{9}{12}$

MPB220

For More Join: @IITJEE_Advanced

Paragraph for Question No. 28 to 29

Two buses A and B are scheduled to arrive at a town central bus station at noon. The probability that bus A will be late is 1/5. The probability that bus B will be late is 7/25. The probability that the bus B is late given that bus A is late is 9/10. Then the probabilities

28. The probability that neither bus will be late on a particular day is

(A) $\frac{2}{25}$ (B) $\frac{18}{25}$ (C) $\frac{7}{10}$ (D) $\frac{12}{25}$

MPB221

29. The probability that bus A is late given that bus B is late, is

(A) $\frac{12}{28}$ (B) $\frac{22}{28}$ (C) $\frac{18}{28}$ (D) $\frac{2}{28}$

MPB222

Paragraph for Question No. 30 to 31

Two whole numbers are randomly selected and multiplied. Consider two events E_1 and E_2 defined as

E_1 : Their product is divisible by 5

E_2 : Unit's place in their product is 5.

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30. Probability $P(E_1)$ is

MPB223

31. Probability $P\left(\frac{E_2}{E_1}\right)$ is

MPB224

Paragraph for Question No. 32 to 33

A uniform unbiased die is constructed in the shape of a regular tetrahedron with faces numbered 2, 2, 3 and 4 and the score is taken from the face on which the die lands. If two such dice are thrown together.

32. Probability of scoring exactly 6 on each of 3 successive throws, can not be

(A) $\left(\frac{2}{16}\right)^3$ (B) $\left(\frac{3}{16}\right)^3$ (C) $\left(\frac{5}{16}\right)^3$ (D) $\left(\frac{6}{16}\right)^3$

MPB225

33. Probability of scoring more than 4 on at least one of the three successive throws, is greater than

(A) $\frac{63}{64}$ (B) $\frac{21}{64}$ (C) $\frac{42}{64}$ (D) $\frac{3}{64}$

MPB226

For More Join: @IITJEE_Advanced

MATCHING LIST TYPE QUESTION

34.

	List-I		**List-II**
(I)	If the probability that units digit in square of an even integer is 4 is p, then the value of $5p$ is	(P)	1
(II)	If A and B are independent events and $P(A \cap B) = \frac{1}{6}$, $P(A) = \frac{1}{3}$, then $6P\left(\frac{B}{\overline{A}}\right) =$	(Q)	2
(III)	One mapping is selected at random from all mappings of the set $S = \{1, 2, 3, ..., n\}$ into itself. If the probability that the mapping is one-one is $\frac{3}{32}$, then the value of n is	(R)	3
(IV)	A boy has 20% chance of hitting at a target. Let p denote the probability of hitting the target for the first time at the nth trial. If p satisfies the inequality $625p^2 - 175p + 12 < 0$, then value of n is	(S)	4

(A) (I) → (Q), (II) → (R), (III) → (S), (IV) → (R)
(B) (I) → (R), (II) → (Q), (III) → (R), (IV) → (S)
(C) (I) → (P), (II) → (S), (III) → (R), (IV) → (Q)
(D) (I) → (Q), (II) → (P), (III) → (Q), (IV) → (R)

MPB227

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MATRIX MATCH TYPE QUESTION

35.

	Column-I		**Column-II**
(A)	Two different numbers are taken from the set {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. The probability that their sum and positive difference, are both multiple of 4, is $x/55$ then x equals	(P)	4
		(Q)	6
(B)	There are two red, two blue, two white and certain number (greater than 0) of green socks in a drawer. If two socks are taken at random from the drawer without replacement, the probability that they are of the same colour is 1/5 then the number of green socks are	(R)	8
(C)	A drawer contains a mixture of red socks and blue socks, atmost 17 in all. It so happens that when two socks are selected randomly without replacement, there is a probability of exactly $1/2$ that both are red or both are blue. The largest possible number of red socks in the drawer that is consistent with this data, is	(S)	10

MPB228

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. There are 2 groups of subjects one of which consists of 5 science subjects & 3 engg. subjects & other consists of 3 science & 5 engg. subjects. An unbiased die is cast . If the number 3 or 5 turns up a subject is selected at random from first group, otherwise the subject is selected from 2nd group. If the probability that an engg. subject is selected is k, then the value of 48k is

MPB229

2. A is one of the 6 horses entered for a race, and is to be ridden by one of two jockeys B or C. It is 2 to 1 that B rides A, in which case all the horses are equally likely to win; if C rides A, his chance is trebled. If the odds against his winning is $\frac{p}{q}$ (Where p & q are coprime), then the value of pq is

MPB230

3. The entries in a two-by-two determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ are integers that are chosen randomly and independently, and, for each entry, the probability that the entry is odd is p. If the probability that the value of the determinant is even is 1/2, then the value of $\left[\frac{1000}{p}\right]$ (where [.] denotes greatest integer function) is

MPB231

4. A room has three electric lamps. From a collection of 10 electric bulbs of which 6 are good 3 are selected at random & put in the lamps. If the probability that the room is lighted λ. Then the value of 90λ is

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MPB232

5. An author writes a good book with a probability of 1/2. If it is good it is published with a probability of 2/3. If it is not, it is published with a probability of 1/4. The probability that he will get atleast one book published if he writes two is k, then the value of 576 k is

MPB233

6. A biased coin which comes up heads three times as often as tails is tossed. If it shows heads, a chip is drawn from urn-I which contains 2 white chips and 5 red chips. If the coin comes up tail, a chip is drawn from urn-II which contains 7 white and 4 red chips. Given that a red chip was drawn. If the probability that the coin came up heads is p. Then the value of $\left[\frac{1}{p}\right]$ (where [.] denotes greatest integer function) is

MPB234

7. A pair of fair dice is tossed. Find the probability that the maximum of the two numbers is greater than 4.

MPB235

8. The chance of one event happening is the square of the chance of a 2nd event, but odds against the first are the cubes of the odds against the 2nd, then the chances of 2nd event happening is (assume that both events are neither sure nor impossible).

MPB236

For More Join: @IITJEE_Advanced

9. Players A and B alternately toss a biased coin, with A going first. A wins if A tosses a Tail before B tosses a Head; otherwise B wins. If the probability of a head is p and the value of p (for which the game is fair to both players) is $\frac{\sqrt{a}-1}{b}$. Then the value of $\frac{a}{b}$ is

MPB237

10. In a batch of 10 articles, 4 articles are defective. 6 articles are taken from the batch for inspection. If more than 2 articles in this batch are defective, the whole batch is rejected Find the probability that the batch will be rejected.

MPB238

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EXERCISE - JEE (Main) PYQ

1. An urn contains 5 red and 2 green balls. A ball is drawn at random from the urn. If the drawn ball is green, then a red ball is added to the urn and if the drawn ball is red, then a green ball is added to the urn; the original ball is not returned to the urn. Now, a second ball is drawn at random from it. The probability that the second ball is red, is : **[JEE (Main) 2019]**

(1) $\frac{26}{49}$ (2) $\frac{32}{49}$ (3) $\frac{27}{49}$ (4) $\frac{21}{49}$

MPB046

2. Two integers are selected at random from the set $\{1, 2, \ldots, 11\}$. Given that the sum of selected numbers is even, the conditional probability that both the numbers are even is : **[JEE (Main) 2019]**

(1) $\frac{2}{5}$ (2) $\frac{1}{2}$ (3) $\frac{3}{5}$ (4) $\frac{7}{10}$

MPB047

3. A bag contains 30 white balls and 10 red balls. 16 balls are drawn one by one randomly from the bag with replacement. If X be the number of white balls drawn, the $\left(\frac{\text{mean of } X}{\text{standard deviation of } X}\right)$ is equal to:- **[JEE (Main) 2019]**

(1) 4 (2) $\frac{4\sqrt{3}}{3}$ (3) $4\sqrt{3}$ (4) $3\sqrt{2}$

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MPB048

4. Four persons can hit a target correctly with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ and $\frac{1}{8}$ respectively. if all hit at the target independently, then the probability that the target would be hit, is **[JEE (Main) 2019]**

(1) $\frac{25}{192}$ (2) $\frac{1}{192}$ (3) $\frac{25}{32}$ (4) $\frac{7}{32}$

MPB049

5. For and initial screening of an admission test, a candidate is given fifty problems to solve. If the probability that the candidate can solve any problem is $\frac{4}{5}$, then the probability that he is unable to solve less than two problems is : **[JEE (Main) 2019]**

(1) $\frac{316}{25}\left(\frac{4}{5}\right)^{48}$ (2) $\frac{54}{5}\left(\frac{4}{5}\right)^{49}$ (3) $\frac{164}{25}\left(\frac{1}{5}\right)^{48}$ (4) $\frac{201}{5}\left(\frac{1}{5}\right)^{49}$

MPB050

6. An unbiased coin is tossed 5 times. Suppose that a variable X is assigned the value k when k consecutive heads are obtained for $k = 3, 4, 5$ otherwise X takes the value -1. Then the expected value of X, is : **[JEE (Main) 2020]**

(1) $\frac{3}{16}$ (2) $-\frac{3}{16}$ (3) $\frac{1}{8}$ (4) $-\frac{1}{8}$

MPB051

For More Join: @IITJEE_Advanced

7. A random variable X has the following probability distribution :

X	:	1	2	3	4	5
$P(X)$	:	K^2	$2K$	K	$2K$	$5K^2$

Then $P(X > 2)$ is equal to : **[JEE (Main) 2020]**

(1) $\frac{7}{12}$ (2) $\frac{23}{36}$ (3) $\frac{1}{36}$ (4) $\frac{1}{6}$

MPB052

8. In a box, there are 20 cards, out of which 10 are libelled as A and the remaining 10 are labelled as B. Cards are drawn at random, one after the other and with replacement, till a second A-card is obtained. The probability that the second A-card appears before the third B-card is : **[JEE (Main) 2020]**

(1) $\frac{11}{16}$ (2) $\frac{13}{16}$ (3) $\frac{9}{16}$ (4) $\frac{15}{16}$

MPB053

9. In a game two players A and B take turns in throwing a pair of fair dice starting with player A and total of scores on the two dice, in each throw is noted. A wins the game if he throws a total of 6 before B throws a total of 7 and B wins the game if he throws a total of 7 before A throws a total of six The game stops as soon as either of the players wins. The probability of A winning the game is : **[JEE (Main) 2020]**

(1) $\frac{31}{61}$ (2) $\frac{5}{6}$ (3) $\frac{5}{31}$ (4) $\frac{30}{61}$

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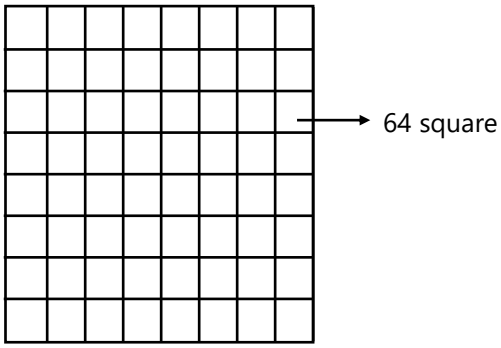
MPB054

10. The probability that a randomly chosen 5-digit number is made from exactly two digits is : **[JEE (Main) 2020]**

(1) $\frac{121}{10^4}$ (2) $\frac{150}{10^4}$ (3) $\frac{135}{10^4}$ (4) $\frac{134}{10^4}$

MPB239

11. Two squares are chosen at random on a chessboard (see figure). The probability that they have a side in common is :



[JEE (Main) 2021]

(1) $\frac{2}{7}$ (2) $\frac{1}{18}$ (3) $\frac{1}{7}$ (4) $\frac{1}{9}$

MPB240

For More Join: @IITJEE_Advanced

12. Let X be a random variable with distribution.

x	-2	-1	3	4	6
$P(X = x)$	$\frac{1}{5}$	A	$\frac{1}{3}$	$\frac{1}{5}$	b

If the mean of X is 2.3 and variance of X is σ^2, then $100\,\sigma^2$ is equal to : **[JEE (Main) 2021]**

MPB055

13. A pack of cards has one card missing. Two cards are drawn randomly and are found to be spades. The probability that the missing card is not a spade, is : **[JEE (Main) 2021]**

(1) $\frac{3}{4}$ (2) $\frac{52}{867}$ (3) $\frac{39}{50}$ (4) $\frac{22}{425}$

MPB056

14. Let A, B and C be three events such that the probability that exactly one of A and B occurs is $(1-k)$, the probability that exactly one of B and C occurs is $(1-2k)$, the probability that exactly one of C and A occurs is $(1-k)$ and the probability of all A, B and C occur simultaneously is k^2, where $0 < k < 1$. Then the probability that at least one of A, B and C occur is : **[JEE (Main) 2021]**

(1) greater than $\frac{1}{8}$ but less than $\frac{1}{4}$ (2) greater than $\frac{1}{2}$

(3) greater than $\frac{1}{4}$ but less than $\frac{1}{2}$ (4) exactly equal to $\frac{1}{2}$

MPB057

15. Four dice are thrown simultaneously and the numbers shown on these dice are recorded in 2×2 matrices. The probability that such formed matrices have all different entries and are non-singular, is : **[JEE (Main) 2021]**

(1) $\frac{45}{162}$ (2) $\frac{23}{81}$ (3) $\frac{22}{81}$ (4) $\frac{43}{162}$

MPB058

16. If the sum and the product of mean and variance of a binomial distribution are 24 and 128 respectively, then the probability of one or two successes is : **[JEE (Main) 2022]**

(1) $\frac{33}{2^{32}}$ (2) $\frac{33}{2^{29}}$ (3) $\frac{33}{2^{28}}$ (4) $\frac{33}{2^{27}}$

MPB059

17. A six faced die is biased such that $3 \times P(a \text{ prime number}) = 6 \times P(a \text{ composite number}) = 2 \times P(1)$. Let X be a random variable that counts the number of times one gets a perfect square on some throws of this die. If the die is thrown twice, then the mean of X is : **[JEE (Main) 2022]**

(1) $\frac{3}{11}$ (2) $\frac{5}{11}$ (3) $\frac{7}{11}$ (4) $\frac{8}{11}$

MPB060

18. Let S be the sample space of all five digit numbers. If p is the probability that a randomly selected number from S, is a multiple of 7 but not divisible by 5, then $9p$ is equal to **[JEE (Main) 2022]**

(1) 1.0146 (2) 1.2085 (3) 1.0285 (4) 1.1521

MPB061

For More Join: @IITJEE_Advanced

19. Let A and B be two events such that $P(B|A)=\frac{2}{5}$, $P(A|B)=\frac{1}{7}$ and $P(A\cap B)=\frac{1}{9}$. Consider $(S1)\,P(A'\cup B)=\frac{5}{6}$, $(S2)\,P(A'\cap B')=\frac{1}{18}$. Then **[JEE (Main) 2022]**

(1) Both $(S1)$ and $(S2)$ are true
(2) Both $(S1)$ and $(S2)$ are false
(3) Only $(S1)$ is true
(4) Only $(S2)$ is true

MPB062

20. Bag I contains 3 red, 4 black and 3 white balls and Bag II contains 2 red, 5 black and 2 white balls. One ball is transferred from Bag I to Bag II and then a ball is draw from Bag II. The ball so drawn is found to be black in colour. Then the probability, that the transferred ball is red, is: **[JEE (Main) 2022]**

(1) $\frac{4}{9}$ (2) $\frac{5}{18}$ (3) $\frac{1}{6}$ (4) $\frac{3}{10}$

MPB063

21. Let N be the sum of the numbers appeared when two fair dice are rolled and let the probability that $N-2,\sqrt{3N},N+2$ are in geometric progression be $\frac{k}{48}$. Then the value of k is **[JEE (Main) 2023]**

(1) 2 (2) 4 (3) 16 (4) 8

MPB107

22. Let $S=\{w_1,w_2,....\}$ be the sample space associated to a random experiment. Let $P(w_n)=\frac{P(w_{n-1})}{2}, n\geq 2$. Let $A=\{2k+3\ell\,;\,k,\ell\in\mathbb{N}\}$ and $B=\{w_n\,;n\in A\}$. Then $P(B)$ is equal to **[JEE (Main) 2023]**

(1) $\frac{3}{32}$ (2) $\frac{3}{64}$ (3) $\frac{1}{16}$ (4) $\frac{1}{32}$

MPB108

23. The urns A, B and C contain 4 red, 6 black; 5 red, 5 black and λ red, 4 black balls respectively. One of the urns is selected at random and a ball is drawn. If the ball drawn is red and the probability that it is drawn from urn C is 0.4 then the square of the length of the side of the largest equilateral triangle, inscribed in the parabola $y^2 = \lambda x$ with one vertex at the vertex of the parabola is **[JEE (Main) 2023]**

MPB111

24. 25% of the population are smokers. A smoker has 27 times more chances to develop lung cancer then a non-smoker. A person is diagnosed with lung cancer and the probability that this person is a smoker is $\frac{k}{10}$. Then the value of k is ____. **[JEE (Main) 2023]**

MPB112

25. In a binomial distribution $B(n, p)$, the sum and product of the mean & variance are 5 and 6 respectively, then find $6(n + p - q)$ is equal to :- **[JEE (Main) 2023]**

(1) 51 (2) 52 (3) 53 (4) 50

MPB113

For More Join: @IITJEE_Advanced

26. If an unbiased die, marked with –2, –1, 0, 1, 2, 3 on its faces, is through five times, then the probability that the product of the outcomes is positive, is : **[JEE (Main) 2023]**

(1) $\frac{881}{2592}$ (2) $\frac{521}{2592}$ (3) $\frac{440}{2592}$ (4) $\frac{27}{288}$

MPB114

27. In a tournament, a team plays 10 matches with probabilities of winning and losing each match as $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Let x be the number of matches that the team wins, and y be the number of matches that team loses. If the probability $P(|x - y| \leq 2)$ is p, then 3^9p equals..... **[JEE (Main) 2024]**

MPB241

28. The coefficients a, b, c in the quadratic equation $ax^2 + bx + c = 0$ are chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. The probability of this equation having repeated roots is : **[JEE (Main) 2024]**

(1) $\frac{3}{256}$ (2) $\frac{1}{128}$ (3) $\frac{1}{64}$ (4) $\frac{3}{128}$

MPB242

29. A company has two plants A and B to manufacture motorcycles. 60% motorcycles are manufactured at plant A and the remaining are manufactured at plant B. 80% of the motorcycles manufactured at plant A are rated of the standard quality, while 90% of the motorcycles manufactured at plant B are rated of the standard quality. A motorcycle picked up randomly from the total production is found to be of the standard quality. If p is the probability that it was manufactured at plant B, then $126p$ is **[JEE (Main) 2024]**

(1) 54 (2) 64 (3) 66 (4) 56

MPB243

30. There are three bags X, Y and Z. Bag X contains 5 one-rupee coins and 4 five-rupee coins; Bag Y contains 4 one-rupee coins and 5 five-rupee coins and Bag Z contains 3 one-rupee coins and 6 five-rupee coins. A bag is selected at random and a coin drawn from it at random is found to be a one-rupee coin. Then the probability, that it came from bag Y, is : **[JEE (Main) 2024]**

(1) $\frac{1}{3}$ (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) $\frac{5}{12}$

MPB244

31. Three balls are drawn at random from a bag containing 5 blue and 4 yellow balls. Let the random variables X and Y respectively denote the number of blue and Yellow balls. If $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively, then $7\bar{X} + 4\bar{Y}$ is equal to _____. **[JEE (Main) 2024]**

MPB245

32. Let Ajay will not appear in JEE exam with probability $p = \frac{2}{7}$, while both Ajay and Vijay will appear in the exam with probability $q = \frac{1}{5}$. Then the probability, that Ajay will appear in the exam and Vijay will not appear is : **[JEE (Main) 2024]**

(1) $\frac{9}{35}$ (2) $\frac{18}{35}$ (3) $\frac{24}{35}$ (4) $\frac{3}{35}$

MPB246

For More Join: @IITJEE_Advanced

33. A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required and let $a = P(X = 3)$, $b = P(X \geq 3)$ and $c = P(X \geq 6 \mid X > 3)$. Then $\frac{b+c}{a}$ is equal to ____.

[JEE (Main) 2024]

MPB247

34. Let the mean and the standard deviation of the probability distribution

X	α	1	0	–3
$P(X)$	$\frac{1}{3}$	K	$\frac{1}{6}$	$\frac{1}{4}$

be μ and σ, respectively. If $\sigma - \mu = 2$, then $\sigma + \mu$ is equal to ____.

[JEE (Main) 2024]

MPB248

35. A coin is based so that a head is twice as likely to occur as a tail. If the coin is tossed 3 times, then the probability of getting two tails and one head is- **[JEE (Main) 2024]**

(1) $\frac{2}{9}$ (2) $\frac{1}{9}$ (3) $\frac{2}{27}$ (4) $\frac{1}{27}$

MPB249

Rankers Academy JEE

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EXERCISE - JEE (Advanced) PYQ

1. Four persons independently solve a certain problem correctly with probabilities $\frac{1}{2},\frac{3}{4},\frac{1}{4},\frac{1}{8}$. Then the probability that the problem is solved correctly by at least one of them is

[JEE (Advanced) 2013]

(A) $\frac{235}{256}$ (B) $\frac{21}{256}$ (C) $\frac{3}{256}$ (D) $\frac{253}{256}$

MPB250

2. Of the three independent events E_1, E_2 and E_3, the probability that only E_1 occurs is α, only E_2 occurs is β and only E_3 occurs is γ. Let the probability p that none of events E_1, E_2 or E_3 occurs satisfy the equations $(\alpha - 2\beta)\,p = \alpha\beta$ and $(\beta - 3\gamma)\,p = 2\beta\gamma$. All the given probabilities are assumed of lie in the interval (0,1). Then $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3} =$ **[JEE (Advanced) 2013]**

MPB251

Paragraph for Question 3 and 4

A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B_3 contains 3 white balls, 4 red balls and 5 black balls.

3. If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other ball is red, the probability that these 2 balls are drawn from box B_2 is

[JEE (Advanced) 2013]

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(A) $\frac{116}{181}$ (B) $\frac{126}{181}$ (C) $\frac{65}{181}$ (D) $\frac{55}{181}$

MPB252

4. If 1 ball is drawn from each of the boxes B_1, B_2 and B_3, the probability that all 3 drawn balls are of the same colour is **[JEE (Advanced) 2013]**

(A) $\frac{82}{648}$ (B) $\frac{90}{648}$ (C) $\frac{558}{648}$ (D) $\frac{566}{648}$

MPB253

5. Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is - **[JEE (Advanced) 2014]**

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

MPB254

Paragraph For Questions 6 and 7

Box 1 contains three cards bearing numbers, 1,2,3 ; box 2 contains five cards bearing numbers 1,2,3,4,5; and box 3 contains seven cards bearing numbers 1,2,3,4,5,6,7. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from the ith box, i = 1,2,3.

6. The probability that $x_1 + x_2 + x_3$ is odd, is - **[JEE (Advanced) 2014]**

(A) $\frac{29}{105}$ (B) $\frac{53}{105}$ (C) $\frac{57}{105}$ (D) $\frac{1}{2}$

MPB255

For More Join: @IITJEE_Advanced

7. The probability that x_1, x_2, x_3 are in an arithmetic progression, is- **[JEE (Advanced) 2014]**

(A) $\frac{9}{105}$ (B) $\frac{10}{105}$ (C) $\frac{11}{105}$ (D) $\frac{7}{105}$

MPB256

8. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96, is **[JEE (Advanced) 2015]**

MPB257

Paragraph For Questions 9 and 10

Let n_1 and n_2 be the number of red and black balls respectively, in box I. Let n_3 and n_4 be the number of red and black balls, respectively, in box II.

9. One of the two boxes, box I and box II, was selected at random and a ball was drawn randomly out of this box. The ball was found to be red. If the probability that this red ball was drawn from box II is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1, n_2, n_3 and n_4 is(are)

[JEE (Advanced) 2015]

(A) $n_1 = 3, n_2 = 3, n_3 = 5, n_4 = 15$ (B) $n_1 = 3, n_2 = 6, n_3 = 10, n_4 = 50$
(C) $n_1 = 8, n_2 = 6, n_3 = 5, n_4 = 20$ (D) $n_1 = 6, n_2 = 12, n_3 = 5, n_4 = 20$

MPB258

10. A ball is drawn at random from box I and transferred to box II. If the probability of drawing a red ball from box I, after this transfer, is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1 and n_2 is(are) **[JEE (Advanced) 2015]**

Bankers Academy JEE

(A) $n_1 = 4$ and $n_2 = 6$ (B) $n_1 = 2$ and $n_2 = 3$
(C) $n_1 = 10$ and $n_2 = 20$ (D) $n_1 = 3$ and $n_2 = 6$

MPB259

11. A computer producing factory has only two plants T_1 and T_2. Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced. 7% of computers produced in the factory turn out to be defective. It is known that

P(computer turns out to be defective given that is produced in plant T_1)
$= 10P$(computer turns out to be defective given that it is produced in plant T_2)

where $P(E)$ denotes the probability of an event E. A computer produces in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is **[JEE (Advanced) 2016]**

(A) $\frac{36}{73}$ (B) $\frac{47}{79}$ (C) $\frac{78}{93}$ (D) $\frac{75}{83}$

MPB260

Paragraph For Questions 12 and 13

Football teams T_1 and T_2 have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of T_1 winning, drawing and losing a game against T_2 are $\frac{1}{2}$, $\frac{1}{6}$ and $\frac{1}{3}$, respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams T_1 and T_2, respectively, after two games

For More Join: @IITJEE_Advanced

12. $P(X > Y)$ is- **[JEE (Advanced) 2016]**

(A) $\frac{1}{4}$ (B) $\frac{5}{12}$ (C) $\frac{1}{2}$ (D) $\frac{7}{12}$

MPB261

13. $P(X = Y)$ is- **[JEE (Advanced) 2016]**

(A) $\frac{11}{36}$ (B) $\frac{1}{3}$ (C) $\frac{13}{36}$ (D) $\frac{1}{2}$

MPB262

14. Let X and Y be two events such that $P(X)=\frac{1}{3}$, $P(X \mid Y)=\frac{1}{2}$ and $P(Y \mid X)=\frac{2}{5}$. Then

[JEE (Advanced) 2017]

(A) $P(X' \mid Y)=\frac{1}{2}$ (B) $P(X \cap Y)=\frac{1}{5}$

(C) $P(X \cup Y)=\frac{2}{5}$ (D) $P(Y)=\frac{4}{15}$

MPB263

15. Three randomly chosen nonnegative integers x, y and z are found to satisfy the equation $x + y + z = 10$. Then the probability that z is even, is **[JEE (Advanced) 2017]**

(A) $\frac{36}{55}$ (B) $\frac{6}{11}$ (C) $\frac{5}{11}$ (D) $\frac{1}{2}$

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MPB264

Paragraph For Questions 16 and 17

There are five students S_1,S_2,S_3,S_4 and S_5 in a music class and for them there are five seats R_1, R_2, R_3, R_4 and R_5 arranged in a row, where initially the seat R_i is allotted to the student S_i, $i = 1, 2, 3, 4, 5$. But, on the examination day, the five students are randomly allotted the five seats.

(*There are two questions based on Paragraph "A". the question given below is one of them*)

16. The probability that, on the examination day, the student S_1 gets the previously allotted seat R_1 and **NONE** of the remaining students gets the seat previously allotted to him/her is -

[JEE (Advanced) 2018]

(A) $\frac{3}{40}$ (B) $\frac{1}{8}$ (C) $\frac{7}{40}$ (D) $\frac{1}{5}$

MPB265

17. For $i = 1, 2, 3, 4$, let T_i denote the event that the students S_i and S_{i+1} do **NOT** sit adjacent to each other on the day of the examination. Then the probability of the event $T_1 \cap T_2 \cap T_3 \cap T_4$ is-

[JEE (Advanced) 2018]

(A) $\frac{1}{15}$ (B) $\frac{1}{10}$ (C) $\frac{7}{60}$ (D) $\frac{1}{5}$

MPB266

For More Join: @IITJEE_Advanced

18. There are three bags B_1, B_2 and B_3. The bag B_1 contains 5 red and 5 green balls, B_2 contains 3 red and 5 green balls, and B_3 contains 5 red and 3 green balls, Bags B_1, B_2 and B_3 have probabilities $\frac{3}{10}$, $\frac{3}{10}$ and $\frac{4}{10}$ respectively of being chosen. A bag is selected at random and a ball is chosen at random from the bag. Then which of the following options is/are correct ?

[JEE (Advanced) 2019]

(A) Probability that the selected bag is B_3 and the chosen ball is green equals $\frac{3}{10}$

(B) Probability that the chosen ball is green equals $\frac{39}{80}$

(C) Probability that the chosen ball is green, given that the selected bag is B_3, equals $\frac{3}{8}$

(D) Probability that the selected bag is B_3, given that the chosen balls is green, equals $\frac{5}{13}$

MPB267

19. Let S be the sample space of all 3 × 3 matrices with entries from the set {0, 1}. Let the events E_1 and E_2 be given by

$E_1 = \{A \in S : \det A = 0\}$ and

$E_2 = \{A \in S :$ sum of entries of A is 7$\}$.

If a matrix is chosen at random from S, then the conditional probability $P(E_1|E_2)$ equals ___

[JEE (Advanced) 2019]

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MPB268

20. Let $|X|$ denote the number of elements in set X. Let $S = \{1,2,3,4,5,6\}$ be a sample space, where each element is equally likely to occur. If A and B are independent events associated with S, then the number of ordered pairs (A, B) such that $1 \leq |B| < |A|$, equals **[JEE (Advanced) 2019]**

MPB269

21. Let C_1 and C_2 be two biased coins such that the probabilities of getting head in a single toss are $\frac{2}{3}$ and $\frac{1}{3}$, respectively. Suppose α is the number of heads that appear when C_1 is tossed twice, independently, and suppose β is the number of heads that appear when C_2 is tossed twice, independently, Then the probability that the roots of the quadratic polynomial $x^2 - \alpha x + \beta$ are real and equal, is **[JEE (Advanced) 2020]**

(A) $\frac{40}{81}$ (B) $\frac{20}{81}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

MPB270

22. The probability that a missile hits a target successfully is 0.75. In order to destroy the target completely, at least three successful hits are required. Then the minimum number of missiles that have to be fired so that the probability of completely destroying the target is NOT less than 0.95, is ____

[JEE (Advanced) 2020]

MPB271

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23. Two fair dice, each with faces numbered 1,2,3,4,5 and 6 are rolled together and the sum of the numbers on the faces is observed. This process is repeated till the sum is either a prime number or a perfect square. Suppose the sum turns out to be a perfect square before it turns out to be a prime number. If p is the probability that this perfect square is an odd number, then the value of $14p$ is ____ **[JEE (Advanced) 2020]**

MPB272

24. Consider three sets $E_1 = \{1, 2, 3\}$, $F_1 = \{1, 3, 4\}$ and $G_1 = \{2, 3, 4, 5\}$. Two elements are chosen at random, without replacement, from the set E_1, and let S_1 denote the set of these chosen elements. Let $E_2 = E_1 - S_1$ and $F_2 = F_1 \cup S_1$. Now two elements are chosen at random, without replacement, from the set F_2 and let S_2 denote the set of these chosen elements. Let $G_2 = G_1 \cup S_2$. Finally, two elements are chosen at random, without replacement, from the set G_2 and let S_3 denote the set of these chosen elements. Let $E_3 = E_2 \cup S_3$. Given that $E_1 = E_3$, let p be the conditional probability of the event $S_1 = \{1, 2\}$. Then the value of p is **[JEE (Advanced) 2021]**

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{1}{2}$ (D) $\frac{2}{5}$

MPB273

Question Stem for Question Nos. 25 and 26

Question Stem

Three numbers are chosen at random, one after another with replacement, from the set S = {1, 2, 3, ..., 100}. Let p_1 be the probability that the maximum of chosen numbers is at least 81 and p_2 be the probability that the minimum of chosen numbers is at most 40. **[JEE (Advanced) 2021]**

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25. The value of $\frac{625}{4}p_1$ is _____.

MPB274

26. The value of $\frac{125}{4}p_2$ is _____.

MPB275

27. Let E, F and G be three events having probabilities $P(E)=\frac{1}{8}, P(F)=\frac{1}{6}$ and $P(G)=\frac{1}{4}$ and let $P(E\cap F\cap G)=\frac{1}{10}$. For any event H, if H^C denotes its complement, then which of the following statements is(are) **TRUE** ? **[JEE (Advanced) 2021]**

(A) $P(E\cap F\cap G^C)\le\frac{1}{40}$ (B) $P(E^C\cap F\cap G)\le\frac{1}{15}$

(C) $P(E\cup F\cup G)\le\frac{13}{24}$ (D) $P(E^C\cap F^C\cap G^C)\le\frac{5}{12}$

MPB276

28. A number is chosen at random from the set {1, 2, 3, ..., 2000}. Let p be the probability that the chosen number is a multiple of 3 or a multiple of 7. Then the value of $500p$ is____. **[JEE (Advanced) 2021]**

MPB277

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29. In a study about a pandemic, data of 900 persons was collected. It was found that
190 persons had symptom of fever,
220 persons had symptom of cough,
220 persons had symptom of breathing problem,
330 persons had symptom of fever or cough or both,
350 persons had symptom of cough or breathing problem or both,
340 persons had symptom of fever or breathing problem or both,
30 persons had all three symptoms (fever, cough and breathing problem).
If a person is chosen randomly from these 900 persons, then the probability that the person has at most one symptom is _____. **[JEE (Advanced) 2022]**

MPB278

30. Two players, P_1 and P_2, play a game against each other. In every round of the game, each player rolls a fair die once, where the six faces of the die have six distinct numbers. Let x and y denote the readings on the die rolled by P_1 and P_2, respectively. If $x > y$, then P_1 scores 5 points and P_2 scores 0 point. If $x = y$, then each player scores 2 points. If $x < y$, then P_1 scores 0 point and P_2 scores 5 points. Let X_i and Y_i be the total scores of P_1 and P_2, respectively, after playing the ith round.

	List-I		**List-II**
(I)	Probability of $(X_2 \geq Y_2)$ is	(P)	$\frac{3}{8}$
(II)	Probability of $(X_2 > Y_2)$ is	(Q)	$\frac{11}{16}$
(III)	Probability of $(X_3 = Y_3)$ is	(R)	$\frac{5}{16}$
(IV)	Probability of $(X_3 > Y_3)$ is	(S)	$\frac{355}{864}$
		(T)	$\frac{77}{432}$

The correct option is: **[JEE (Advanced) 2022]**

(A) (I) → (Q); (II) → (R); (III) → (T); (IV) → (S)
(B) (I) → (Q); (II) → (R); (III) → (T); (IV) → (T)
(C) (I) → (P); (II) → (R); (III) → (Q); (IV) → (S)
(D) (I) → (P); (II) → (R); (III) → (Q); (IV) → (T)

MPB279

31. Suppose that
Box-I contains 8 red, 3 blue and 5 green balls,
Box-II contains 24 red, 9 blue and 15 green balls,
Box-III contains 1 blue, 12 green and 3 yellow balls,
Box-IV contains 10 green, 16 orange and 6 white balls.
A ball is chosen randomly from Box-I ; call this ball b. If b is red then a ball is chosen randomly from Box-II, if b is blue then a ball is chosen randomly from Box-III, and if b is green then a ball is chosen randomly from Box-IV. The conditional probability of the event 'one of the chosen balls is white' given that the event 'at least one of the chosen balls is green' has happened, is equal to

[JEE (Advanced) 2022]

(A) $\frac{15}{256}$ (B) $\frac{3}{16}$ (C) $\frac{5}{52}$ (D) $\frac{1}{8}$

MPB280

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32. Let $X=\left\{(x, y) \in \mathbb{Z} \times \mathbb{Z}: \frac{x^2}{8}+\frac{y^2}{20}<1 \text{ and } y^2<5x\right\}$. Three distinct points P, Q and R are randomly chosen from X. Then the probability that P, Q and R form a triangle whose area is a positive integer, is **[JEE (Advanced) 2023]**

(A) $\frac{71}{220}$ (B) $\frac{73}{220}$ (C) $\frac{79}{220}$ (D) $\frac{83}{220}$

MPB281

33. Consider an experiment of tossing a coin repeatedly until the outcomes of two consecutive tosses are same. If the probability of a random toss resulting in head is $\frac{1}{3}$, then the probability that the experiment stops with head is. **[JEE (Advanced) 2023]**

(A) $\frac{1}{3}$ (B) $\frac{5}{21}$ (C) $\frac{4}{21}$ (D) $\frac{2}{7}$

MPB282

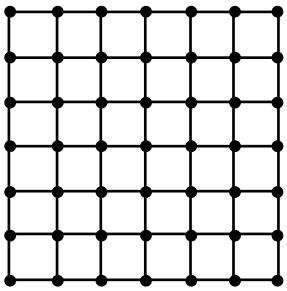
34. Let X be the set of all five digit numbers formed using 1,2,2,2,4,4,0. For example, 22240 is in X while 02244 and 44422 are not in X. Suppose that each element of X has an equal chance of being chosen. Let p be the conditional probability that an element chosen at random is a multiple of 20 given that it is a multiple of 5. Then the value of $38p$ is equal to **[JEE (Advanced) 2023]**

MPB283

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PARAGRAPH "II"

Consider the 6 × 6 square in the figure. Let $A_1, A_2, \ldots, A_{49}$ be the points of intersections (dots in the picture) in some order. We say that A_i and A_j are friends if they are adjacent along a row or along a column. Assume that each point A_i has an equal chance of being chosen.



(There are two questions based on PARAGRAPH "II", the question given below is one of them)

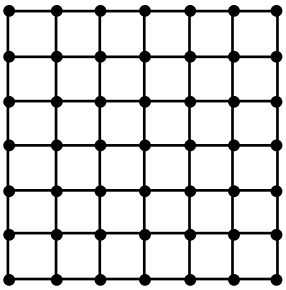
35. Let p_i be the probability that a randomly chosen point has i many friends, i = 0,1,2,3,4. Let X be a random variable such that for i = 0,1,2,3,4, the probability $P(X = i) = p_i$. Then the value of $7E(X)$ is **[JEE (Advanced) 2023]**

MPB284

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PARAGRAPH "II"

Consider the 6 × 6 square in the figure. Let $A_1, A_2, \ldots, A_{49}$ be the points of intersections (dots in the picture) in some order. We say that A_i and A_j are friends if they are adjacent along a row or along a column. Assume that each point A_i has an equal chance of being chosen.



(There are two questions based on PARAGRAPH "II", the question given below is one of them)

36. Two distinct points are chosen randomly out of the points $A_1, A_2, \ldots, A_{49}$. Let p be the probability that they are friends. Then the value of $7p$ is **[JEE (Advanced) 2023]**

MPB285

37. A student appears for a quiz consisting of only true-false type questions and answers all the questions. The student knows the answers of some questions and guesses the answers for the remaining questions. Whenever the student knows the answer of a question, he gives the correct answer. Assume that the probability of the student giving the correct answer for a question, given that he has guessed it, is $\frac{1}{2}$. Also assume that the probability of the answer for a question being guessed, given that the student's answer is correct, is $\frac{1}{6}$. Then the probability that the student knows the answer of a randomly chosen question is : **[JEE (Advanced) 2024]**

(A) $\frac{1}{12}$ (B) $\frac{1}{7}$ (C) $\frac{5}{7}$ (D) $\frac{5}{12}$

MPB286

38. A bag contains N balls out of which 3 balls are white, 6 balls are green, and the remaining balls are blue. Assume that the balls are identical otherwise. Three balls are drawn randomly one after the other without replacement. For $i = 1, 2, 3$, let W_i, G_i, and B_i denote the events that the ball drawn in the ith draw is a white ball, green ball, and blue ball, respectively. If the probability $P(W_1 \cap G_2 \cap B_3) = \frac{2}{5N}$ and the conditional probability $P(B_3 \mid W_1 \cap G_2) = \frac{2}{9}$, then N equals ___

[JEE (Advanced) 2024]

MPB287

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	C	A	D	B	C	A	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	A	A	C	A	C	C	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	B	B	D	D	D	A	A	C

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C,D	A,B,C,D	A,B,D	A,C,D	B,C,D	A,B	B,C	A,D	A,C,D	A,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,D	B,D	A,B	A,D	C,D	A,B	A,B,C	B,D	A,B,C,D	A,D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A,B,C,D	A,B	A,B	A,C,D	A,B	C	B	C	C	0.36
Que.	31	32	33	34	35					
Ans.	0.25	A,B,D	B,C,D	A	A → Q, B → P, C → S					

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EXERCISE (S)

1.	26	**2.**	65	**3.**	1414	**4.**	87	**5.**	407
6.	1.00	**7.**	0.55	**8.**	0.33	**9.**	2.50	**10.**	0.45

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	3	3	2	3	2	1	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	781	3	2	4	3	4	3	1	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	432	9	2	2	8288	3	1	1
Que.	31	32	33	34	35					
Ans.	17	2	12	5	1					

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EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	6	D	A	A	B	C	8	A,B	C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	C	A,D	B	A	C	B,C	0.50	422
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	6	8.00	A	76.25	24.50	A,B,C	214	0.80	A
Que.	31	32	33	34	35	36	37	38		
Ans.	C	B	B	31	24.00	0.50	C	11		

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